INCH-POUND

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17 May 1993
SUPERSEDING
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31 January 1985

MILITARY SPECIFICATION

SKID BASE, WOOD, FOR UNITIZATION OF BOXED AMMUNITION

This specification is approved for use by the US Army Armament Research and Development Center and is available for use by all Departments and agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the material, design, and construction of 4-way entry skid bases. The type I and II skid bases have a maximum load capacity of 2,200 pounds. Exceptions to the maximum load capacity must be approved by the proponent of this specification.

1.2 Classification. Skid base units covered by this specification shall be of the following types:

Type I - Skid base, plywood deck, variable dimension, 4-way entry, see paragraph 3 and Figures 1 and 2.

Type II - Skid base, wing type, variable dimension, 4-way entry, see paragraph 3 and Figures 3 and 4

1.2.1 Size. The length and width of the skid base will be calculated in accordance with the "special notes" sections on Figures 2 and 4 or as specified in the contractual document. The length will be the dimension running parallel to the runners on a Type I skid base and perpendicular to the runners on a Type II skid base.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-208	- Ink, Marking, Stencil, Opaque (Porous and non-Porous Surfaces)
A-A-55057	- Panels, Wood/Wood Based: Construction and Decorative
FF-F-133	- Fastener, Corrugated, Wood Joint (Saw Edge)
FF-N-105	- Nails, Brads, Staples and Spikes: Wire Cut and Wrought

MILITARY

MIL-A-48078	- Ammunition Quality Assurance Special Provisions

STANDARDS

MILITARY

MIL STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL STD-109	- Quality Assurance Terms and Definitions
MIL STD-129	- Marking for Shipment and Storage
MIL STD-731	- Quality of Wood Members for Containers and Pallets
MIL STD-1168	- Lot Numbering of Ammunition
MIL STD-1235	- Single and Multi-Level Continuous Sampling Procedures and Table for Inspection by Attributes

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

American Society for Testing and Materials (ASTM):

ASTM D-3953	- Standard Specification for Strapping, Flat Steel and Seals
ASTM D-4442	- Test Methods for Direct Moisture Content Measurement of Wood and Wood Base Material
ASTM D-4444	- Test Methods for Use and Calibration of Hand Held Moisture Meters
ASTM D-4689	- Standard Specification for Adhesive, Casein Type

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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Voluntary Product Standard:

PS 56-73 - Structural Glued Laminated Timber

(Application for copies should be addressed to the American Institute of Timber Construction, 333 West Hampden Avenue, Englewood, CO 88110.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 Design and construction. Design and construction will be as specified in Table I and Figures 1, 2, 3, and 4 as applicable.

3.2 Materials.

3.2.1 Wood. All wood shall be nominal size except as indicated in figures 1 through 4 and shall conform to MIL STD-731, A-A-55057, OR PS 56-73 as applicable. Group woods of a species listed in Tables II or III will be used as specified in Table I for skid base parts. The plywood deck and alternative parts for Type I skid base shall be 3/4 inch plywood, Group B, PS-1, Grade CD interior with exterior glue (see Table I). If specified grade is not available, a better grade may be substituted (Tables II and III).

3.2.1.1 Quality of lumber. The individual pieces of lumber will be free of the defects defined below:

a. Presence of Decay.

b. Warp, bow, cup or crook-pieces of lumber in an assembly which provides an uneven bearing surface or in themselves vary from a true or plane surface more than 1/8 inch per linear foot.

c. Divergence in the grain of a piece of lumber in a unit which exceeds one inch in ten inches or less.

d. Wane (bark or lack of wood from any cause) on more than one edge or exceeding 3/4 inch in width. If oak or chestnut are used, the wood or wood products must be absolutely free from bark.

e. Holes (worm holes, knotholes, unsound knots or other) the sum of widths of which exceed 1//3 the width of the piece.

f. Sound knots exceeding three inches in width or exceeding 1/3 the width of the piece.

g. Insect and beetle infestation which could result in subsequent unit destruction.

h. Chipped or torn grain or machine skid of more than 1/8-inch vibration and more than the width of the piece in length.

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3.2.1.1 Quality of lumber cont'd

i. Posts with shakes or checks greater than 1-1/2 inches in depth and/or running over 1/2 the length of the post, are defective unless properly repaired.

j. Unacceptable lumber imperfections: Streaks or pockets of pitch, strain or machine burns which interfere with the marking requirements.

TABLE I. Skid base.

Type I

		Material			
Name of part	Key No.	Kind		Grade	Spec No.
Plywood Deck	3 & 3A	Group B (PS-1 (Interior With Exterior Glue	1/	CD	A-A-55057
Post A & B	1 & 2	Wood, Class 2 Structural	1/	Group II or III (see Table II)	MIL-STD-731
Post A & B Alternative	1A & 2A	Group B (PS-1) (Interior with Exterior Glue)	1/	CD	A-A-55057
Crosstie	4	Wood, Class 2 Structural	1/	Group II or III (see Table II	MIL-STD-731
Crosstie Alternative	4A	Group B (PS-1) (Interior with Exterior Glue)	1/	CD	A-A-55057
Runner	6	Wood, Class 2 Structural	1/	Group II or III (see Table II)	MIL-STD-731
Runner Alternative	5A	Group B (PS-1) (Interior with Exterior Glue)	1/	CD	A-A-55057
Filler Block	5	Wood, Class 2 Structural	1/	Group II or III (See Table II)	MIL-STD-731
Adhesive					ASTM D 4689

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TABLE I. Skid base. (continued)

Type I

		Material			
Name of Part	Key No.	Kind		Grade	Spec No.
Nails	3, 3A 5A & 6	See Key No.'s			FF-N-105
Top Deck Board	3 & 4	Wood class 2 Structural	1/	Group I, II or III (see Table II	MIL-STD-731
Post	1	Wood class 2 Structural	1/	Group II or III (see Table II)	MIL-STD-731
Stringer Board	2	Wood class 2 Structural	1/	Group II or III (see Table II	MIL-STD-731
Runner	5	Wood, class 2 Structural	1/	Group III (see Table II)	MIL-STD-731
Nails	3, 4 & 5	See Key No.s			FF-N-105

1/ - Laminated wood, Groups I, II and III of Table III may be substituted.

TABLE II. Wood groups.

Group I	Group II	Group III
Tamarack	Elm	Hickory
Hemlock, western	Ash (except white)	Pecan
Fir, silver	Pine, southern yellow	Elm, rock
Pine, alaska	Gum	Birch, sweet or yellow
Fir, douglas	Maple, soft	Hackberry
Hemlock, eastern	Larch, western	Maple, hard
Magnolia	Fir, douglas (coast)	Ash, white
Cedar	Sycamore	Beech
Aspen	Tupelo, water	Locust
Jeffrey pine		Oak
Spruce		
Chestnut		
Cottonwood		
Fir, white		
Pine, ponderosa		

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TABLE III. Laminated wood.

Group I	Group II	Group III
Fir, douglas	Pine, southern	Maple, sugar
Jeffrey pine	Maple, sugar	Oak, red
Aspen	Larch, western	Oak, white
Pine, lodgepole		

3.2.1.2 Moisture content. At the time of fabrication at the manufacturer's plant, the moisture content of the lumber, including plywood, used in construction of the unit shall not exceed 19 percent maximum for all 1-1/8 inch or less lumber and 26 percent maximum for all lumber exceeding 1-1/8 inch. The testing of moisture content will be conducted in accordance with ASTM D-4442 and ASTM D-4444 as outlined in paragraph 4.3.3.1 immediately prior to preservative treatment.

3.2.2 Fasteners.

3.2.2.1 Corrugated fasteners. Corrugated fasteners will be 3/8 inch for repair of runners, stringers, deck pieces and crossties and will be 3/4 inch for repair of posts. Corrugated fasteners shall conform to the requirements of FF-F-133.

3.2.2.2 Nails. Nails shall be as specified in figures 1 and 3 and will conform to the requirements of FF-N-105.

3.2.3 Skid base component details.

3.2.3.1 Deck boards. Deck boards on a Type II skid base shall be surfaced on the outside faces of the skid base. All deck boards shall be surfaced to a uniform thickness and end deck boards shall be free of wane on their outer edges.

3.2.3.2 Stringers and crossties. Stringers and crossties for a Type I skid base shall be surfaced on at least one side to a uniform thickness.

3.2.3.3 Posts. Posts shall be finished or smooth sawn on top and bottom to a uniform height. Outer faces of all outside posts shall be finished. Ends of posts shall be cut square with sides.

3.2.3.4 Runners. Runners shall be surfaced on the outside faces of the skid base. All runners shall be surfaced to a uniform thickness and shall be free of wane on their outer edges.

3.2.3.5 Component tolerances. Tolerances shall be as specified in figures 1 through 4. Minimum allowable thickness' of softwood lumber from Groups I and II in Table II shall be restricted to the dimensional limitations of MIL-STD-731 for one inch nominal lumber.

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3.3 Assembled skid bases. Assembled skid bases shall be free from the following defects:

a. Dimensional requirements not in conformance with applicable figures or contractual agreement.

b. Nailing requirements not in accordance with applicable figures.

c. Splits:

(1) Splits in posts caused by nailing are not allowed if less than two nails are secure and/or the post has more than one split. Any split in the posts exceeding two inches in depth and/or extending the full length of the post shall be considered defective.

(2) Splits in crossties, runners, stringers and deck pieces that exceed four inches in depth.

3.3.1 Acceptable repairs.

a. Posts. Splits less than two inches in depth and/or less than the full length of the piece may be repaired by use of additional nails or fasteners (see figure 6). Posts five inches or less in length require two fasteners located one inch from each end of the check or shake. Posts over five inches in length require an additional fastener located approximately midway of the shake or check. Fasteners shall be flush with or slightly below the surface of the piece.

b. Crossties, runners, stringers and deck pieces. Splits to the open end through more than one nail may be repaired by nailing (figure 6) provided the nailing reduces the original crack to a hairline and the lumber is secured under the nail heads. End fasteners will be positioned not less than one inch from the end of the deck or deckboards and not less than one inch from the side edges of the deck or deck boards.

3.3.2 Fabrication of skid bases. Skid bases shall be assembled with fasteners specified in 3.2.2 and the requirements of figures 1 through 4. Nails shall be driven in a staggered pattern. Splits caused by nailing shall be supported by additional fasteners (Figure 6). End fasteners shall be positioned not less than 3/4 inch from the end of the deck or deckboards. Edge fasteners shall be positioned not less than one inch from side edges of the deck or deckboards. The remaining fasteners shall be uniformly spaced between them. All fasteners bent in driving shall be removed or broken off below the surface and replaced. The heads of the fasteners shall be driven down beneath the surface of the board.

3.4 Preservative treatment. Assembled skid bases shall be treated with the same water repellant wood preservative specified for the boxes which are to make up the unit load.

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3.4.1 Identification of preservative treated pallets. Treated units shall be identified in the same manner specified for the boxes which are to be unitized on the skid base except that alphabetical designators shall be marked on the outside of the center posts when the box is identified with an alphabetical designator on the bottom of the box.

3.5 First article. Requirements for the submission of the first article shall be as specified in the contract.

3.6 Workmanship. Assembled units shall be clean and properly assembled. Surfaces shall be smooth and free of damage. Marking shall be clear and legible. Nails bent during driving shall be removed or broken off below the surface end and replaced. Nail heads shall be flush to 1/16 inch below the surface. Assembled skid bases shall form a rectangle. All intermediate deck boards on a type II skid base shall be parallel to the edges. Stringer boards, crossties, and posts shall not protrude on any side.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspection set forth herein, or in referenced specifications, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.2 First article inspection.

4.2.1 Submission. Prior to initiating regular production, the contractor shall submit one (1) complete assembly and one (1) complete set of parts to a Government approved facility as designated by the Contracting Officer for evaluation in accordance with 4.2.2.

4.2.1.1 Sample. The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and manufactured using the same production processes, procedures, and equipment that will be used in fulfilling the contract. The first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government.

4.2.1.2 Prior contractor inspection. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects shall not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed. Each assembly and component part shall be examined for every dimension and requirement of the applicable drawing, 4.3.2, 4.3.3, and MIL-A-48078.

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4.2.3 Rejection. If the assembly or any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon observing any nonconformance of the sample to the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation. The term "inspection lot" as used in this specification is defined as an essentially homogenous collection of units of product from which representative samples are drawn and inspected to determine conformance with applicable requirements. The sample selected shall not represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105.

4.3.2. Examination. The sample unit shall be one skid base. Sampling plans and procedures for skid base procurement (see following inspection sample for boxed ammunition), shall be in accordance with MIL-STD-105. The inspection level shall be II. At the option of the procuring activity (see 6.2.1.), AQL's and sample plans may be applied to individual characteristics listed, using an AQL of 0.65 percent for each minor and an AQL of 0.40 percent for each major defect.

Assembled skid base, 4 way for boxed ammunition

MAJOR CATEGORIES	DEFECTS	METHOD OF INSPECTION	CODE NO. See 6.3	AQL
101	Runner slots (type I skid base) obstructed or missing	Visual	01001	2.50
102	Fasteners not per specification (Paragraph 3.2.2)	Visual/Scale	01002	2.50
103	Incorrect species of wood or wood group (Table II).	Visual	01003	2.50
104	Evidence of insect infestation (Paragraph 3.2.1.1.g).	Visual	01004	2.50
105	Decay or excessive worm holes (Paragraph 3.2.1.1.e).	Visual	01005	2.50
106	Nails or nailing pattern not in conformance with the applicable drawing (Figures 1 and 3).	Visual	01006	2.50

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Assembled sid base, 4 way for boxed ammunition defects (Cont'd)

MAJOR CATEGORIES	DEFECTS	METHOD OF INSPECTION	CODE NO. SEE 6.3)	AQL
107	No preservative (when preservative is required (Para 3.4).	Visual/Test	01007	2.50
108	Incorrect plywood group or grade (Table III).	Visual	01008	2.50
109	Any defect or combination of defects occurring in a position that impairs strength of the pallet.	Visual/Scale	01009	2.50
110	Checks, splits and shakes (Paragraph 3.2 and 3.3).	Visual	01010	2.50

MINOR CATEGORIES	DEFECTS	METHOD OF INSPECTION	CODE NO. See 6.3)	AQL
201	Wane (Para 3.2.1.1.d)	Visual/Scale	01011	6.50
202	Not square or accurately formed (Table I and Figures 1, 2, 3, and 4).	Visual/Scale	01012	6.50
203	Warp, bow or crook (Para 3.2.1.1.b).	Visual/Scale	01013	6.50
204	Knots (Para 3.2.1.1.f).	Visual/Scale	01014	6.50
205	Divergence in grain (Para 3.2.1.1.c).	Visual/Scale	01015	6.50
206	Incorrect dimensions (Figures 1, 2, 3, and 4).	Scale	01016	6.50
207	Poor workmanship (Para 3.6).	Visual	01017	6.50
208	Incorrect fasteners for particular use and/or fastener pattern (Para 3.2.2).	Visual/Scale	01018	6.50
209	Marking missing, illegible or incorrect (Para 3.2.1.1.j).	Visual	01019	6.50

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4.3.3 Testing.

4.3.3.1 Moisture content. The wood members of a sample of 15 skid bases from each lot of skid bases shall be tested for moisture content using an electric meter complying with specifications ASTM D-4442 and ASTM D-4444. A minimum of three readings from each sample shall be taken at random locations on the runners, crossties, stringers, deck and posts. The mean value of these readings shall meet the applicable requirements of this specification. Failure to meet these requirements will reject the lot. The mean value of the posts shall be separate from that of the remainder of the skid base. in case of dispute the oven drying method of ASTM D-4442 and ASTM D-4444 shall be used as a verification. For the oven test, five random samples shall be cut from each of the wood members and the determined moisture content shall constitute the value for acceptance. The contractor shall not be reimbursed for material destroyed during the test. Rejected lots destined for Government use shall not be subjected to the requirements of 3.4 without prior approval of the Contracting Officer. These lots may be held in a favorable environment until such time as a retest shows compliance with 3.2.1.2.

4.3.3.2 Presence of preservative. A sample of skid bases shall be randomly selected from each lot and subjected to the tests in paragraph 4.4.2. Five (5) members of each type I Skid Base (one plywood deck, one runner, one stringer, one post and one crossties) shall be subjected to this test. Four (4) members of each Type II Skid Base (one top deckboard, one stringer, one runner, and one post) shall be subjected to this test. If one or more members fail to meet the requirements specified in paragraph 4.4.2, the lot will be rejected.

4.3.3.3 Freedom from solvent (treated skid bases). A sample of 15 skid bases shall be randomly selected from each lot subjected to the tests in paragraph 4.4.3 when a freedom from solvent test is required by the applicable box specification. Five (5) members of each Type I Skid Base (one plywood deck, one runner, one stringer, one post, and one crosstie) shall be subjected to this test. Four (4) members of each Type II Skid Base (one top deckboard, one stringer, one runner, and one post) shall be submitted to this test. If one or more members fail to meet the requirements of paragraph 4.4.3 the lot shall be rejected.

4.3.3.4 Rejected lot. In the event the original sample fails to satisfactorily pass the presence of preservative and freedom from solvent test, the contractor may rework or recondition the rejected lot and resubmit this lot for acceptance. The retest shall be performed using a sample of 30 skid bases randomly selected from the rejected lot. If one or more members fail to meet the requirements specified in paragraphs 4.4.2 or 4.4.3 the lot will be rejected.

4.4 Test methods and procedures.

4.4.1 Moisture content. Moisture content of the lumber shall be determined in accordance with ASTM D-4442 and ASTM D-4444. Moisture content shall be as specified in 3.2.1.2.

4.4.2 Presence of preservative. The same test used to verify the presence of preservative in the box being unitized shall be used to verify the presence of preservative in the skid base (see paragraph 6.2).

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4.4.3 Freedom from solvent. The same test used to verify the freedom from solvent in the box being unitized shall be used to verify the freedom from solvent in the skid base (see paragraph 6.2).

5 PACKAGING

5.1 Unit pack. Not applicable

5.2 Packing. Type I skid bases will be unitized together in an upright position in stable stacks of a maximum of 10. Type II skid bases will be unitized in stable stacks of a maximum of 15 (see figure 5).

5.3 Shipment. Unless otherwise specified by the procuring agency, assembled units will be shipped in carload or truckload lots when quantity permits. shipments shall be offered in such a manner as to insure acceptance by common or other carrier for safe transportation at the lowest rate to point of delivery. The loading of assembled skid bases in railcars or trucks shall be in such a manner as to facilitate handling by forklift equipment. Marking for shipment will be in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The skid bases described herein are for use with ammunition commodities, in general, materials handling operations in the military storage and distribution system: wherever full 4-way entry is Skid bases may be loaded wherever materials enter the supply system for storage purposes and may be shipped loaded from storage. Skid bases may be used for long-term storage in both covered and open storage areas. 6.2

ORDERING DATA.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type of skid base.

c. Title, number and date of the specification which describes the boxes which are to make up the unit load.

d. AQL individual characteristics, if applied (see 4.3.2)

6.3 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.4 Subject term (key word) listing

Bow
Crosstie
Moisture
Post
Runner
Skid base
Wane
Warp

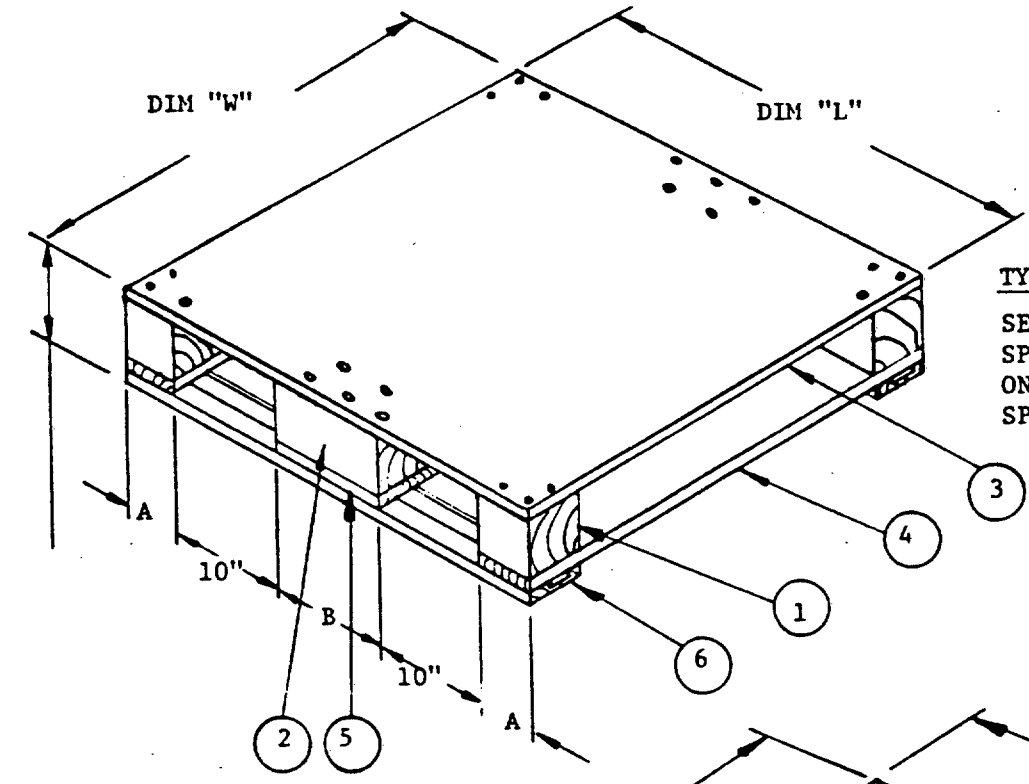
6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue do to the extensiveness of the changes.

Custodians:
Army - AL

Preparing Activity:
Army - AL

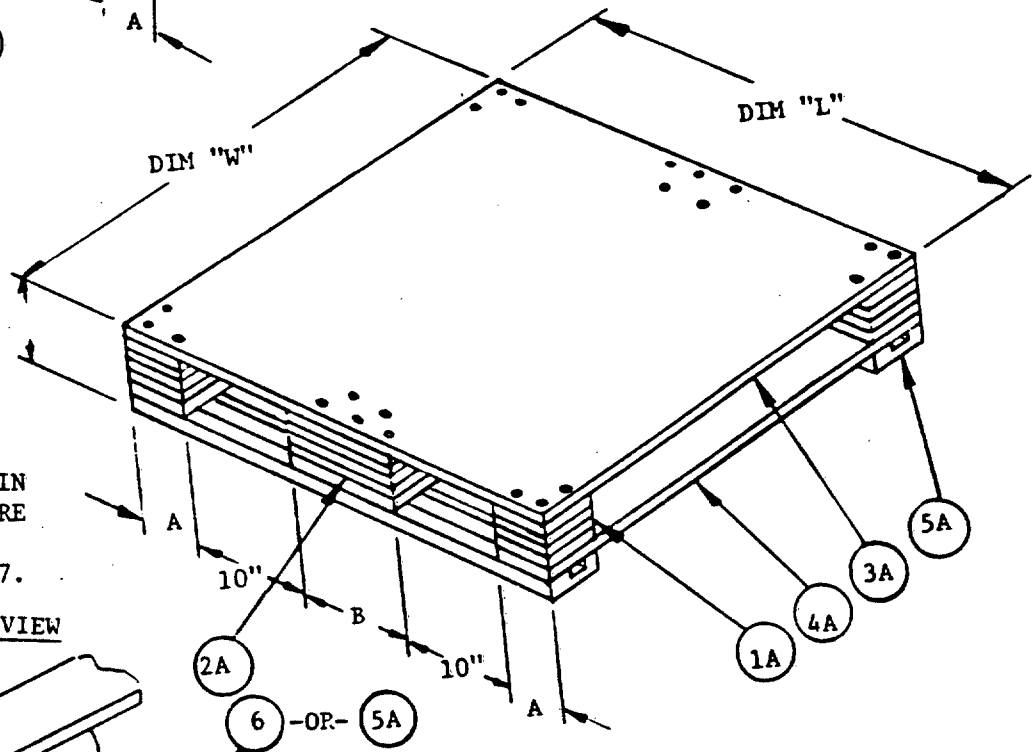
Project Number PACK-A381

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TYPE I SKID BASE

SEE KEY NUMBERS AT RIGHT, SPECIAL NOTES, AND DETAILS ON PAGE 17 AND MATERIAL SPECIFICATIONS ON PAGE 4.



STRAP SHOULD BE STAPLED IN PLACE IF POSITIONED BEFORE RUNNER IS INSTALLED, SEE SPECIAL NOTE 3 ON PAGE 17.

PARTIAL BOTTOM VIEW

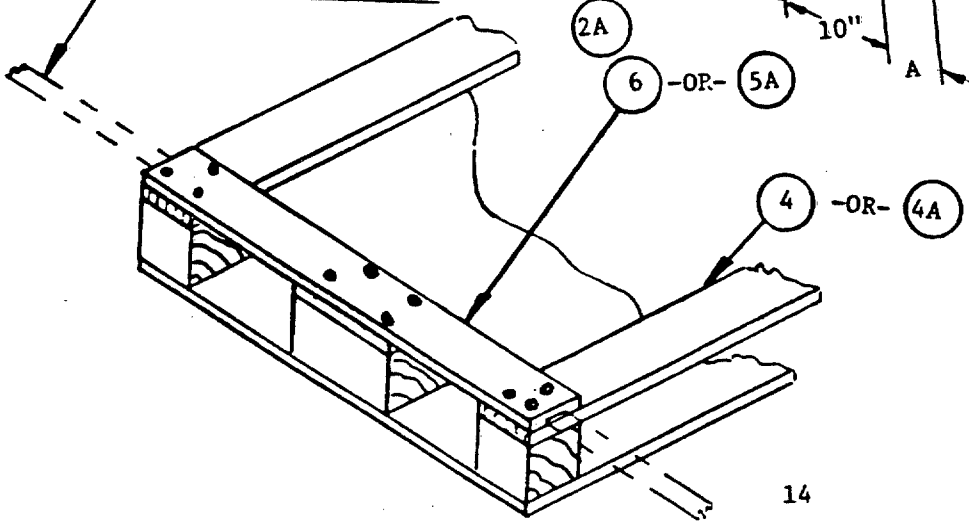


FIGURE I

TYPE I SKID BASE

CONSTRUCTION PROCEDURES

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KEY NUMBERS

1. Post A, 4" X 4" material (4 required). See "Post A" detail and special note 5 on page 17.
2. Post B, 4" X 4" by cut to suit (2 required). See "Post B" detail and special notes 4 and 5 on page 17.
3. Deck, plywood, 3/4" by "Dim L" and "Dim W" to suit (1 required). Nail to Post A W/3-8d Nails (Type II - Style 10) each post and to Post B W/5-8d Nails (Type II - Style 10) each post using staggered nailing pattern shown. See special notes 1 and 2 on page 17.
4. Skid cross tie, 1" X 4" by "Dim W" (2 required). See partial bottom view at left, skid cross tie detail and special notes 1 and 6 on page 17.
5. Filler block, 1" X 4" cut to suit (2 required). See "Filler Block" detail and special notes 4, 5 and 6 on page 17.
6. Runner, 1" X 4" by "Dim L" (2 required). Position as shown and nail thru skid cross tie into Post "A" W/3 (Type II - Style 18) 3" long X .120 diameter. Nails and thru filler block into Post B W/4 (same type) nails using the staggered nailing pattern shown. Do not nail thru strap slot. See Runner detail and special notes 2, 3 and 6 from page 17.

KEY NUMBERS (ALTERNATIVE)

1A. Post A (alternative) plywood, 3 1/2" wide X 3 1/2" long X 3 1/2" high (4 required). Laminate from 1/2" to 1" thick pieces to a height of 3 1/2". See "Post A (alternative)" detail on page 17 and material specification for adhesive on page 4.

2A. Post B (alternative) plywood, 3 1/2" high X 4 1/4" wide by Dim B (2 required). Laminate from 1/2" to 1" thick pieces to a height of 4 1/4". See "Post B (alternative)" detail and special notes 4 and 5 on page 17 and material specifications on page 4 for adhesive.

3A. Deck, plywood, 3/4" by "Dim L" and "Dim W" to suit (1 required). Nail to Post A W/3-8d nails (Type II - Style 10) each post and to Post B W/5-8d nails (Type II - Style 10) each post using staggered nailing pattern shown. See special notes 1 and 2 on page 17.

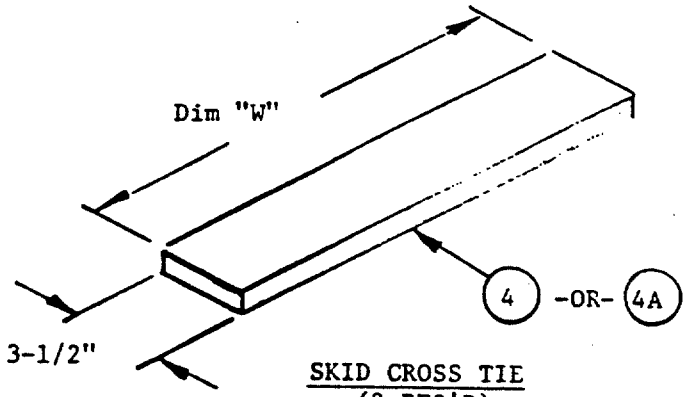
4A. Skid cross tie (alternative) plywood, 3/4" X 3 1/2" by "Dim W" (2 required). See partial bottom view at left, skid cross tie detail and special notes 1 and 6 on page 17.

5A. Runner (alternative) plywood, 3/4" X 3 1/2" by "Dim L" (2 required). Position as shown and nail thru skid cross tie into Post "A" W/3 (Type II - Style 18) 3" long X .120 diameter nails and into Post "B" W/4 (same type) nails using the staggered nailing pattern shown. Do not nail thru strap slot. See runner detail and special notes 2, 3 and 6 on page 17.

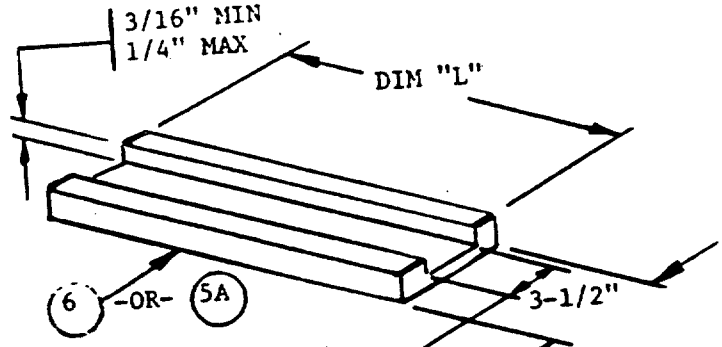
(SPECIAL NOTES)

1. When the skid bases are procured without the runner applied, the cross ties and filler block will be fastened to each post with one 8d common nail. The runners will be nailed to the posts as indicated, to complete a skid base.
2. Skid bases that are procured from a manufacturer will have the under side of the deck stenciled in 1/2" (one-half inch) minimum letters as follows:
   a. Manufacturer's name.
   b. Manufacturer's address.
   c. Date (month and year) of manufacture.
   d. Manufacturer's lot number.
3. Stenciling shall be in accordance with TT-I-1795.

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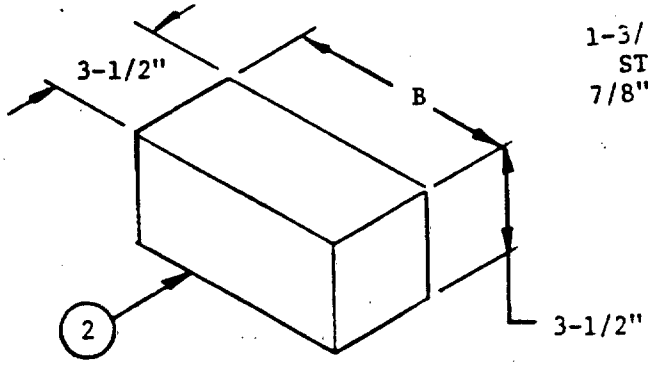


SKID CROSS TIE
(2 REQ'D)

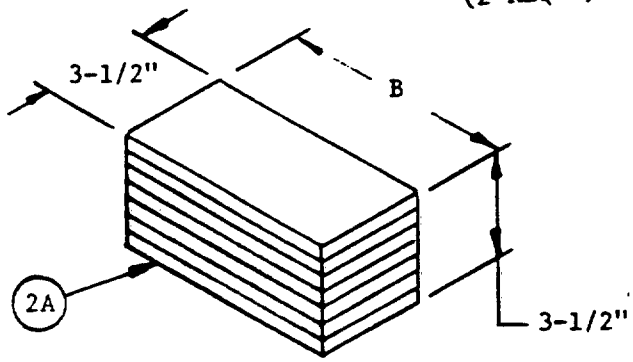


1-3/8" MIN, 1-1/2" MAX FOR 1-1/4" STRAP
7/8" MIN, 1" MAX FOR 3/4" STRAP

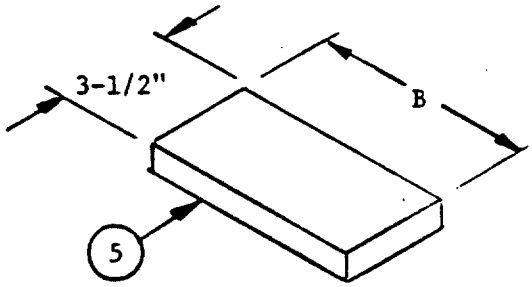
RUNNER
(2 REQ'D)



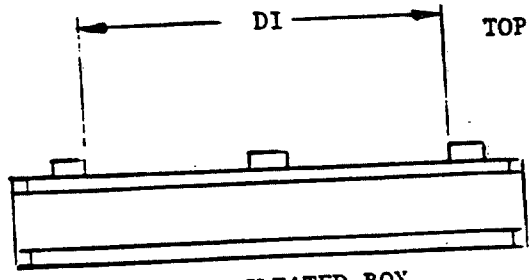
POST B
(2 REQ'D) 4" X 4" MATL.



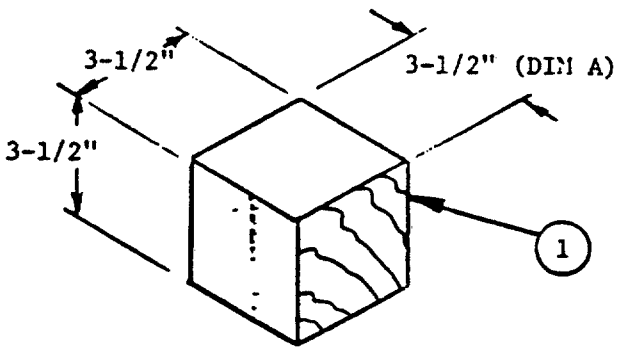
POST B (ALTERNATIVE)
(2 REQ'D) PLYWOOD MATERIAL



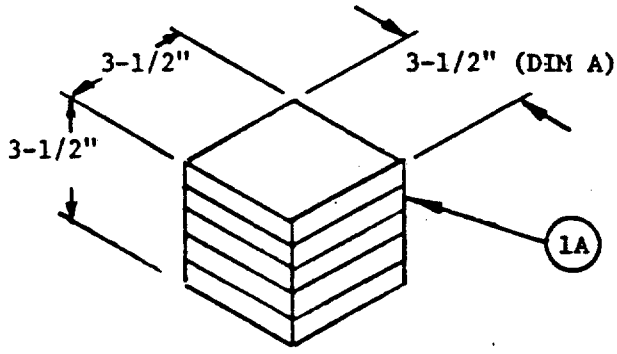
FILLER BLOCK
(2 REQ'D) 1" X 4" MATL.



CLEATED BOX
DI = DISTANCE BETWEEN TWO OUTSIDE TOP CLEATS



POST A
(4 REQ'D) 4" X 4" MATL



POST A (ALTERNATIVE)
(4 REQ'D) PLYWOOD MATERIAL

FIGURE 2 DETAILS (TYPE I SKID BASE)

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SPECIAL NOTES

1. Dimension "W" will be determined as follows:

   a. For cleated boxes more than 38" long, dimension "W" is the distance between the two outside top cleats on the top of the box, minus 1". See "Cleated Box" detail.

   b. For cleated boxes 34" through 38" long, with 25" or more between outside top cleats, dimension "W" will be 24".

   c. For cleated boxes 27" to 34" long, the criteria of 1.a above will be applied (distance between cleats, minus 1").

   d. For wood boxes more than 38" long and without top cleats, the "W" dimension will be one-fifth less than the long dimension of the box. E.g. 45" + 5 = 9", 45" - 9" = Dimension "W" 36".

   e. For wood boxes 27" through 38" long without top cleats, dimension "W" will be 24".

2. Dimension "L" of the skidded base will be the short dimension of the box X the number of boxes per layer on the skidded base minus 1" to provide a skid of slightly lesser dimension than the load length.

3. Runners need not be dadoed if applied after straps are in place.

4. Skidded bases for unit loads with a length dimension 32" or more will have material length dimensions as follows:

   Posts "A" = 3-1/2".
   Posts "B" = Dim "L" - 27".
   Filler block = Dim "L" - 27".

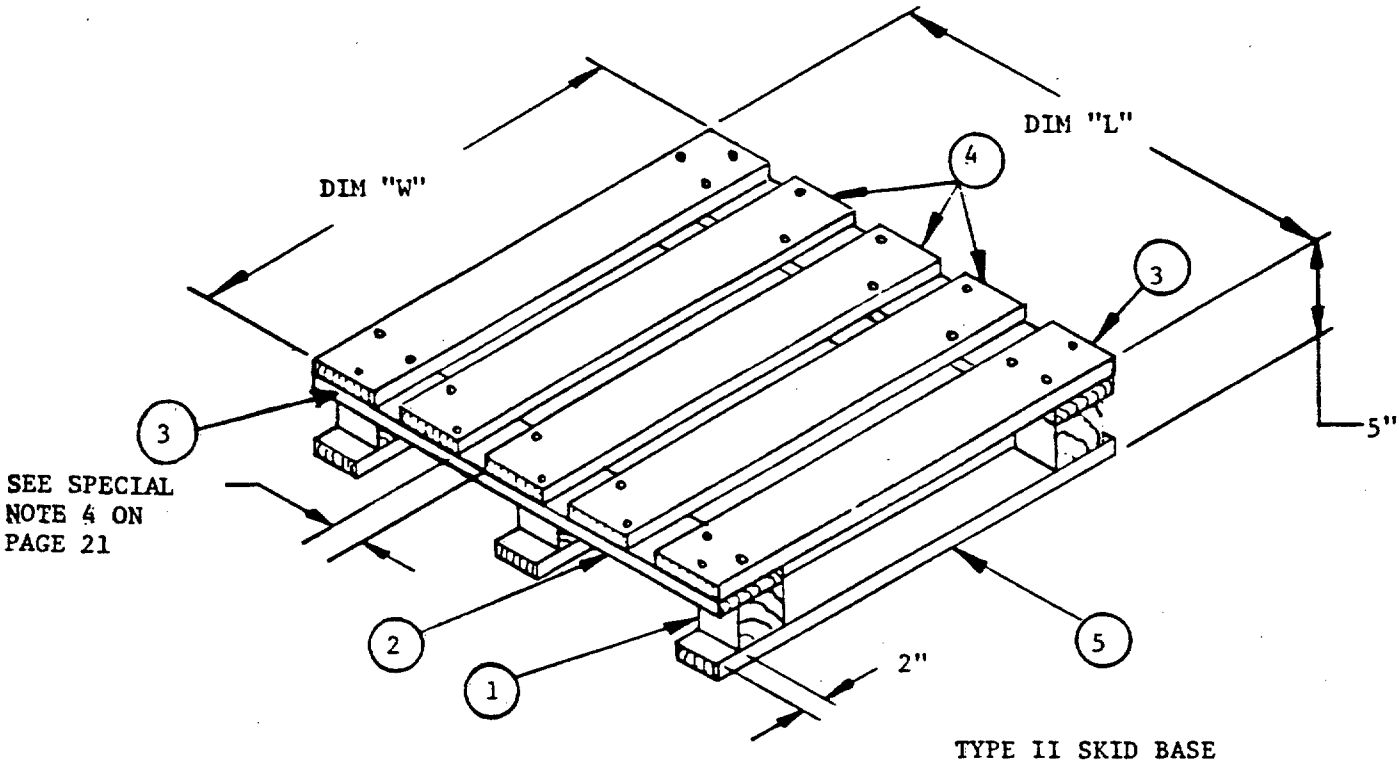
5. Skidded bases for unit loads with a length dimension less than 32" will have fork openings 8" wide and have material length dimensions as follows:

   Posts "A" will be 3-1/2".
   Posts "B" = Dim "L" - 23".
   Filler block = Dim "L" - 23".

   Nailing Pattern and number of nails for Posts "B" will be the same as shown for Posts "A". See top and bottom views on Figure 1.

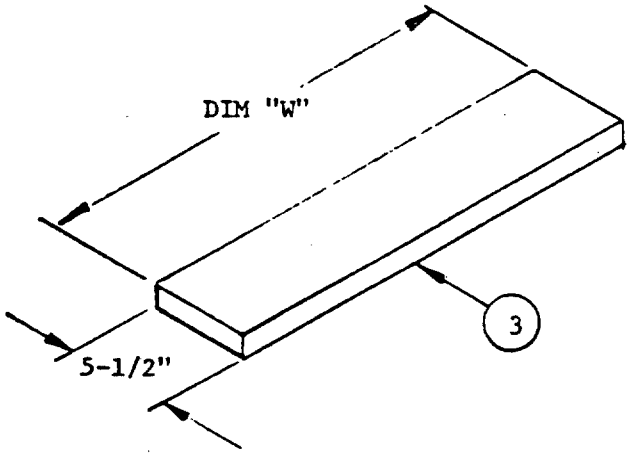
6. When the skid bases are procured completely assembled, the cross-tie, filler block and runner may be fastened in one operation using nails for nailing pattern specified.

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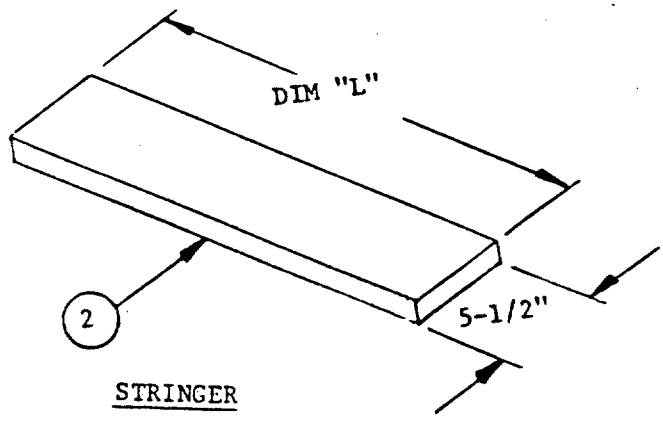
TYPE II SKID BASE

SEE SPECIAL NOTES ON PAGE 21 AND KEY NUMBERS BELOW AND MATERIAL SPECIFICATION ON PAGE 4.



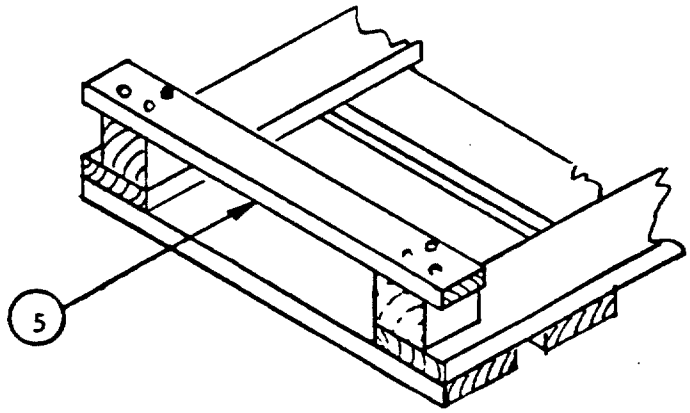
TOP DECK BOARD

(2 REQ'D) 1" X 6" MATERIAL



STRINGER

(2 REQ'D) 1" X 6" MATERIAL



PARTIAL BOTTOM VIEW

FIGURE 3 TYPE II SKID BASE CONSTRUCTION PROCEDURES

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KEY NUMBERS

1. Post 4" wide x 2-3/4" high x 3-1/2" long (6 required). See "Post" detail on page 20.

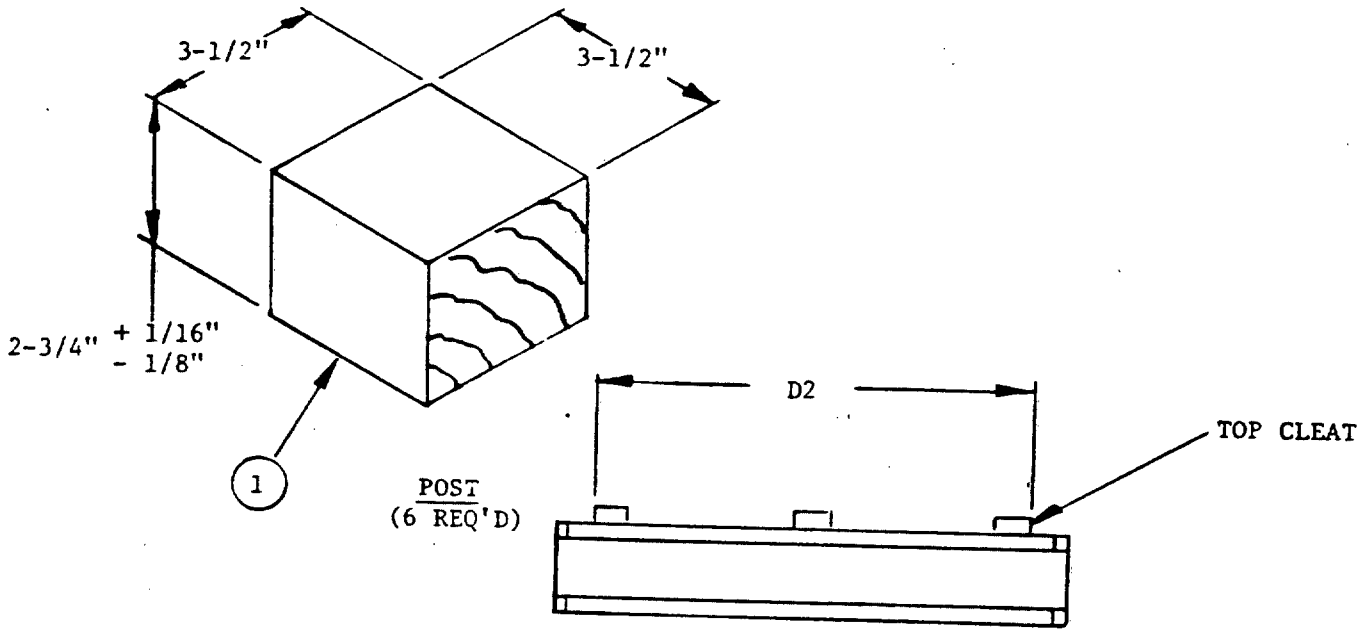
2. Stringer, 1" x 6" by "Dim L" (2 required). Position as shown. See "stringer" detail, special note 1.b. on page 21 and material specifications on page 4.

3. Top deck board, 1" x 6" by "Dim W" (2 required). Position as shown and nail through stringer piece into Post W/3 nails (Type II - Style 18), 3" to 3-1/4" long x .135" diameter at each end with a staggered nailing pattern. See "Top Deck Board" detail and special note 1.a on page 21.

4. Top deck board, 1" material x variable width by "Dim W" (as required). May be 1" x 4" or 1" x 6" material. Position as shown and nail to stringer piece w/2 nails for 4" material and 3 nails for 6" material, (Type II - Style 18), 1-1/2" long x .120 diameter at each end. NOTE: Use 3" to 3-1/4" long x .135" dia nails on center board to penetrate the center posts. See "Top Deck Board" detail on page 20 and special notes 1.a, 4 and 5 on page 21.

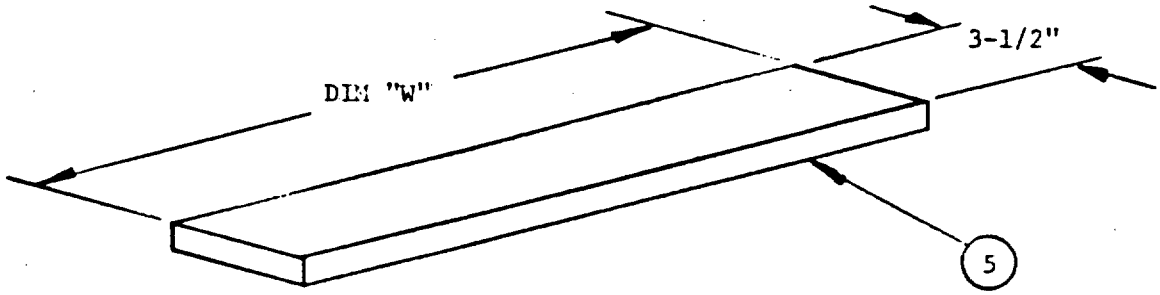
5. Runner, 1" x 4" by "Dim W" (3 required). Position as shown and nail to post w/3 nails (Type II - Style 18), 2-1/4" long x .120" diameter at each joint. See nailing pattern in "Partial Bottom View", runner detail and special note 1.a on page 21.

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CLEATED BOX

D2 = DISTANCE FROM OUTSIDE TO OUTSIDE OF TWO OUTSIDE TOP CLEATS.



RUNNER
(3 REQ'D) 1" X 4" MATERIAL

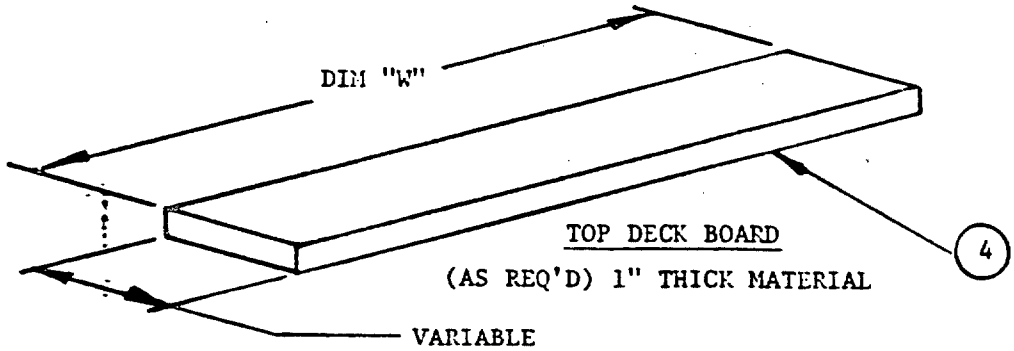


FIGURE 4 DETAILS (TYPE II SKID BASE)

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SPECIAL NOTES

1. Dimensions "W" and "L" will be determined as follows:

   a. Dimension "W" (width of skid base) will be the distance measured outside to outside of the two outside cleats on the top of the box plus 2-1/2" ± 1/8". When unitizing boxes without top cleats, dimension "W" will be one-fifth (1/5) less than the long dimension (box length) of the box: e.g., if the box length is 45", then 45" x 1/5 = 9"; therefore, dimension "W" = 45" - 9" = 36.

   b. Dimension "L" (length of the skid base) will be the narrow dimension (box width) of the box times the number of boxes in the first layer of the skidded unit minus 1" maximum.

2. Unless otherwise specified, only Type II skid bases will be used for unitizing wirebound boxes. Dimension "W" for a type II skid base to be loaded with wirebound boxes will be the long dimension (box length) of the wirebound box minus 1" maximum. This will provide a skid base of a slightly lesser width dimension than the wirebound box length. When strapping a Type II skid base loaded with wirebound boxes and four tiedown straps are required, a tiedown strap will be placed adjacent to the inboard side of each post and a tiedown strap will be placed adjacent to the outboard side of each post. When strapping a Type II skid base loaded with wirebound boxes and only two tiedown straps are required, a tiedown strap will be placed adjacent to the outboard side of each post.

3. Skid bases that are procured from a manufacturer will have the bottom of one outer deck board stenciled in 1/2" (one-half inch) minimum letters as follows:

   a. Manufacturer's name.

   b. Manufacturer's address.

   c. Date (month and year) of manufacture.

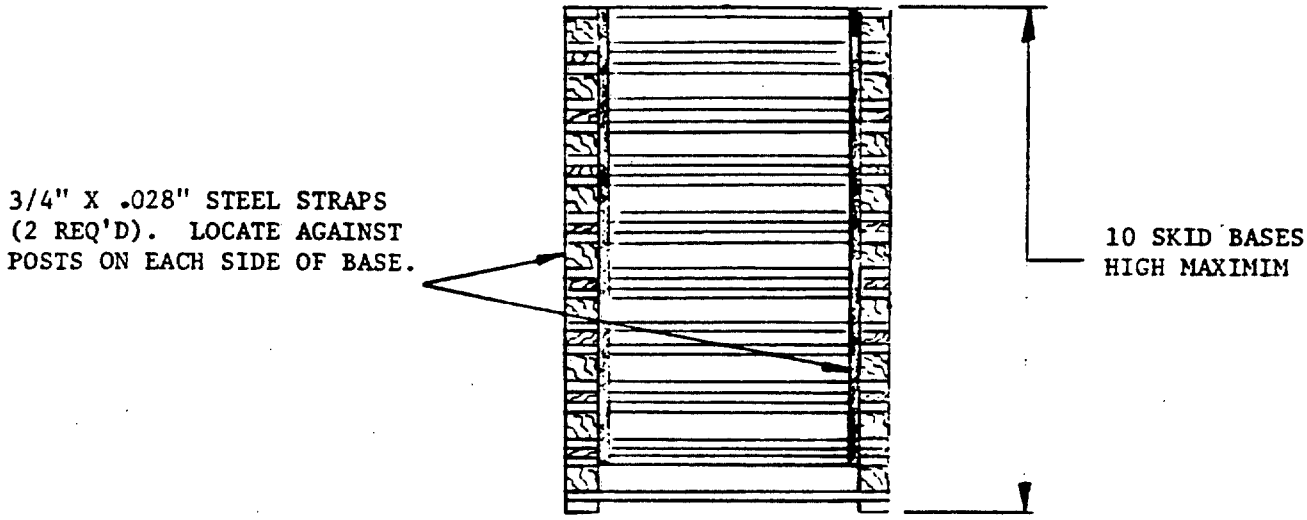
   d. Manufacturer's lot number.

4. The quantity of pieces, "4", as well as the width of the piece may vary depending upon Dimension "L" of the skid base. However, the spacing between top deck boards will be minimum 1" - maximum 3".

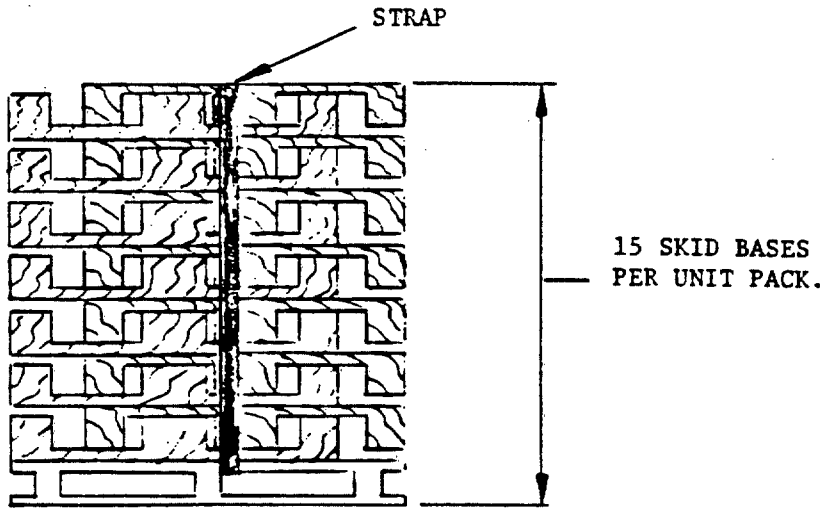
5. For securing top deck boards to stringers, it is permissable to use nail, box, Type II, Style 4A, 1 3/4 inch long, .080 diameter (14 gage) and/or staple, galvanized, 1-5/8 inch long by 3/4 inch crown, 14 gage, clinched (utilizing 30 degree - 50 degree stitches) as alternates to the nail specified on page 18.

6. Stenciling shall be in accordance with TT-I-1795.

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METHOD OF PREPARING TYPE 1 SKID BASES FOR SHIPMENT



METHOD OF PREPARING TYPE II SKID BASES FOR SHIPMENT.

1. PLACE 1 SKID BASE IN NORMAL POSITION.
2. NEST 2 SKID BASES AND POSITION 7 PAIRS AS SHOWN.
3. STRAP USING ONE 3/4" X .028 STEEL STRAP CONFORMING TO TYPE 1, GRADE 1, FINISH C, CLASS 1 OF QQ-S-781.
4. STEEL STRAPPING SHALL BE 3/4" X .028" CONFORMING TO TYPE 1, GRADE 1, FINISH C, CLASS 1 OF QQ-S-781.

FIGURE 5 PREPARATION FOR SHIPMENT

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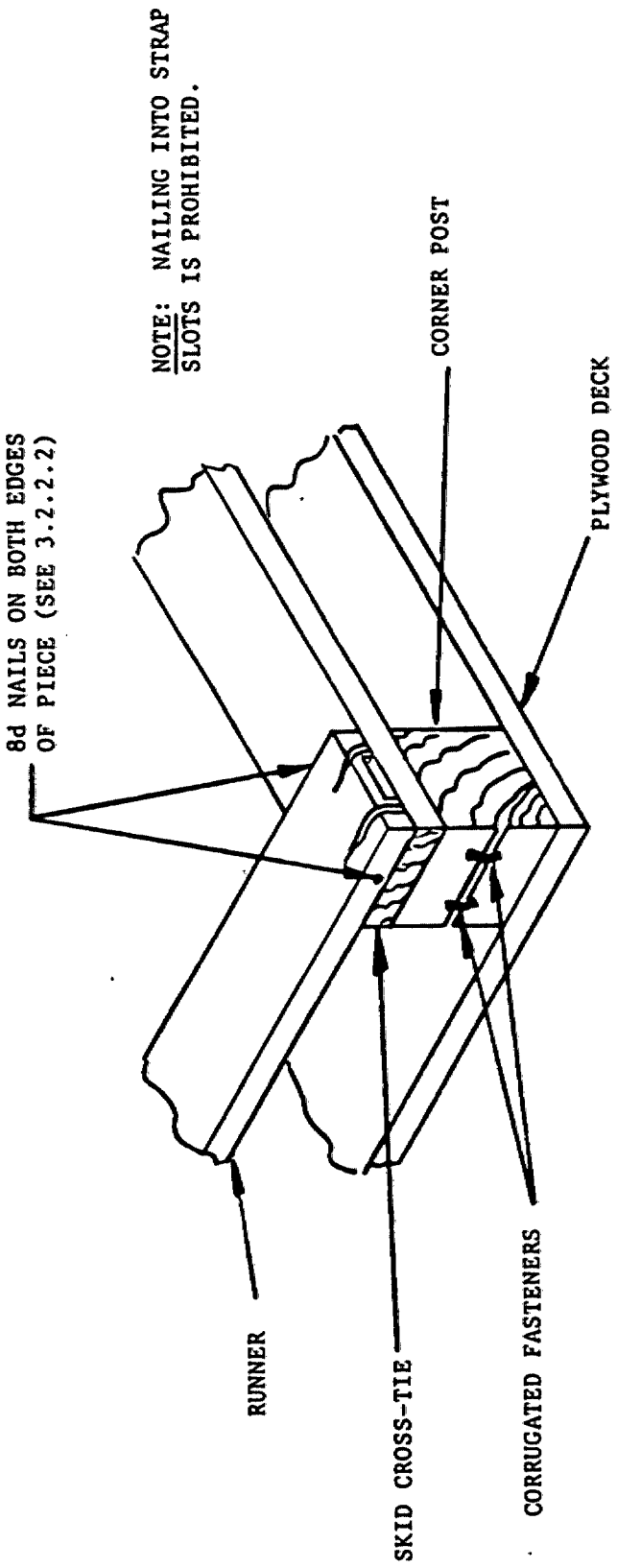


FIGURE 6 ACCEPTABLE METHODS OF REPAIRING SPLITS, CHECKS OR SHAKES

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RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-50786G	2. DOCUMENT DATE (YYMMDD) 930517

3. DOCUMENT TITLE SKID BASE, WOOD, FOR UNITIZATION OF BOXED AMMUNITION

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME DALE HOLLAND	b. TELEPHONE (Include Area Code) (1) Commercial: 309-782-8146 (2) AUTOVON: 793-8146
c. ADDRESS (Include Zip Code) COMMANDER US ARMY ARMAMENT RESEARCH, DEV & ENGR CTR ATTN: SMCAR-EST-P ~~ROCK ISLAND, IL 61299-7300~~	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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